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Convolution theorem for fractional cosine-sine transform and its application

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Fractional cosine transform (FRCT) and fractional sine transform (FRST), which are closely related to the fractional Fourier transform (FRFT), are useful mathematical and optical tool for signal processing. Many properties for these transforms are well investigated, but the convolution theorems are still to be determined. In this paper, we derive convolution theorems for the fractional cosine transform (FRCT) and fractional sine transform (FRST) based on the four novel convolution operations. And then, a potential application for these two transforms on designing multiplicative filter is presented.

Keywords: Fractional Fourier transform; fractional cosine and sine transform; convolution operation; convolution theorem

1. Introduction

The fractional Fourier transform (FRFT) [1–3], which is a generalization of the Fourier transform (FT) [4], has been found as one of the most useful mathematical and optical tools for signal processing. In the past decade, FRFT has attracted much attentions in signal processing community, as the generalization of FT, the relevant theory for FRFT has been developed, including the convolution theorem [5–11], uncertainty principle [3, 12, 13], sampling theory [14–18], and so on.

The fractional cosine transform (FRCT) and fractional sine transform (FRST) [19–21], which are useful signal processing tools, was defined based on the fractional operations' decomposition and eigenfunction. This is similar to the derivation process of the fractional Fourier transform (FRFT) introduced in [1]. Many applications for the fractional cosine transform (FRCT) and the fractional sine transform (FRST) have been found in filter design, optical system analysis, pattern recognition [20], solving differential equations etc.

Some properties for FRCT and FRST are already known, including additivity property scaling, shift, modulation, symmetry relation and so on [21]. However, for the best of our knowledge, the convolution theorems related to the FRST and FRCT are still to be determined. Since the convolution operation is fundamental in information processing, for example, the filter, noise reduction, encryption and watermarking etc. So, it is theoretically interesting and practically useful to consider the fractional cosine transform (FRCT) and fractional sine transform (FRST) involving their convolution theorems.

In this paper, we define four novel convolution operations for FRCT and FRST, the relevant convolution theorems are derived. And the potential applications of the derived results in filter design are also considered. The paper is organized as follows. Section 2 reviews the definition for FRCT and FRST and the important relationships between the FRFT, FRCT and the FRST. Section 3 gives four different convolution operations for FRCT and FRST and derives four different kinds of convolution theorems for FRCT and FRST, the properties and relations of convolution operations are also studied in details. Potential applications of the new convolution theorems for FRCT and FRST are discussed in Section 4 and conclusions are made in Section 5.

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2. Preliminaries

2.1. Convolution theorems for Fourier cosine and sine transform

In 1934, Paley and Wiener proposed the Fourier cosine transforms (FCT) and the Fourier sine transforms (FST) [22], which are defined as follows respectively

$$(F_c f)(s) = F_c(f(t))(s) = \sqrt{\frac{1}{2\pi}} \int_0^{\infty} f(t)\cos(st)\mathrm{d}t,\ s > 0 \tag{2.1}$$

And

$$(F_s f)(s) = F_s(f(t))(s) = \sqrt{\frac{1}{2\pi}} \int_0^{\infty} f(t)\sin(st)\mathrm{d}t,\ s > 0 \tag{2.2}$$

Here, $f(x)$ is continuous and belongs to $L(\mathbf{R}_+)$, where $L(\mathbf{R}_+)$ denote the set of all positive real functions, such that $\int_0^{\infty} |f(t)|\mathrm{d}t < 1$.

Then, the convolution of two functions f and g for the FCT was given by [23]

$$(f \underset{F_c}{\star} g)(t) = \sqrt{\frac{1}{2\pi}} \int_0^{\infty} f(\tau)\left(g(|t-\tau|) + g(t+\tau)\right)\mathrm{d}\tau, \tag{2.3}$$

Which satisfies the following convolution theorem

$$F_c\left[(f \underset{F_c}{\star} g)(t)\right](s) = (F_c f)(s)\,(F_c g)(s),\ s > 0 \tag{2.4}$$

Afterwards, the convolutions for FCT and FST were studied extensively in many papers. For example, convolution for the FCT-FST was introduced by [24]

$$(f \underset{2}{\star} g)(t) = \sqrt{\frac{1}{2\pi}} \int_0^{\infty} f(\tau)\left(g(t+\tau) + \mathrm{sign}(\tau - t)g(|t-\tau|)\right)\mathrm{d}\tau, \tag{2.5}$$

Which satisfies the convolution theorem for FCT-FST

$$F_c\left[(f \underset{2}{\star} g)(t)\right](s) = (F_s f)(s)\,(F_s g)(s),\ s > 0 \tag{2.6}$$

A convolution with the weight function $\gamma(\tau) = \sin\tau$ of functions f and g for the FST was studied in [25, 26]:

$$\begin{aligned}(f \overset{\gamma}{\underset{F_s}{\star}} g)(t) = \frac{1}{2\sqrt{2\pi}} \int_0^{\infty} & f(\tau)\left[g(t+\tau+1) + \mathrm{sign}(t-\tau+1)g(|t-\tau+1|)\right. \\ & \left. + \mathrm{sign}(t+\tau-1)g(|t+\tau-1|) + \mathrm{sign}(t-\tau-1)g(|t-\tau-1|)\right]\mathrm{d}\tau,\end{aligned} \tag{2.7}$$

For which the convolution theorem holds

$$F_s\left[(f \overset{\gamma}{\underset{F_s}{\star}} g)(t)\right](s) = \sin s\,(F_s f)(s)\,(F_s g)(s),\ s > 0 \tag{2.8}$$

Another the convolution with the weight function $\gamma(\tau) = \cos\tau$ of functions f and g for the FCT was studied in [27]

$$(f \overset{\gamma}{\underset{F_c}{\star}} g)(t) = \frac{1}{2\sqrt{2\pi}} \int_0^{\infty} f(\tau)\left[g(t+\tau+1) + g(|t-\tau+1|) + g(|t+\tau-1|) + g(|t-\tau-1|)\right]\mathrm{d}\tau, \tag{2.9}$$

For which the convolution theorem holds

$$F_c\left[(f \overset{\gamma}{\underset{F_c}{\star}} g)(t)\right](s) = \cos s\,(F_c f)(s)\,(F_c g)(s),\ s > 0 \tag{2.10}$$

Others convolution operation and convolution theorem associated with FCT or FST are listed in Table I.

In this paper, based on the research results of fractional Fourier transform, we give four novel convolution operations for the fractional cosine transform (FRCT) and fractional sine transform (FRST). The properties of these newly defined operations are investigated in detail. And the relationships between our novel convolution operations and the well-known convolutions are obtained. The applications of the derived results in filter design are also considered.

Table I. Others convolution operation and convolution theorem.

Tansform	Convolution operation	Convolution theorem
FST-FCT	$(f \underset{1}{\star} g)(t) = \sqrt{\frac{1}{2\pi}} \int_0^\infty f(\tau)$ $\cdot (g(\lvert t-\tau \rvert) - g(t+\tau))\,\mathrm{d}\tau$ (see in [23, 24, 28])	$F_s\left[(f \underset{1}{\star} g)(t)\right](s) = (F_s f)(s)(F_c g)(s), s > 0$ (see in [[23, 24, 28]])
FST-FCT	$(f \overset{\gamma}{\underset{F_s,F_c}{\star}} g)(t) = \frac{1}{2\sqrt{2\pi}} \int_0^\infty f(\tau)$ $\cdot [g(\lvert t+\tau-1 \rvert) + g(\lvert t-\tau-1 \rvert)$ $-g(t+\tau+1) - g(\lvert t-\tau+1 \rvert)]\,\mathrm{d}\tau$ (see in [29])	$F_s\left[(f \overset{\gamma}{\underset{F_s,F_c}{\star}} g)(t)\right](s) = \sin s\,(F_c f)(s)(F_c g)(s), s > 0$ (see in [29])
FCT-FST	$(f \overset{\gamma}{\underset{F_c,F_s}{\star}} g)(t) = \frac{1}{2\sqrt{2\pi}} \int_0^\infty f(\tau)$ $\cdot [g(\lvert t+\tau-1 \rvert) + g(\lvert t-\tau+1 \rvert)$ $-g(t+\tau+1) - g(\lvert t-\tau-1 \rvert)]\mathrm{d}\tau$ (see in [30])	$F_s\left[(f \overset{\gamma}{\underset{F_c,F_s}{\star}} g)(t)\right](s) = \sin s\,(F_s f)(s)(F_c g)(s), s > 0$ (see in [30])

2.2. The fractional Fourier cosine and sine transform

We first review the definition of the fractional Fourier transform (FRFT) [1–3]

$$(F^\alpha f)(s) = F^\alpha(f(t))(s) = \int_{-\infty}^{\infty} f(t) K^\varphi(t,s) e^{-jst\csc\varphi}\,\mathrm{d}t \tag{2.11}$$

The FRFT kernel $K_F^\varphi(t,s) = K^\varphi(t,s)e^{-jst\csc\varphi}$, in which $K^\varphi(t,s)$ is given by

$$K^\varphi(t,s) = \begin{cases} A_\varphi e^{j\left(\frac{t^2+s^2}{2}\right)\cot\varphi}, & \varphi \neq k\pi \\ \delta(t-s), & \varphi = 2k\pi \\ \delta(t+s), & \varphi = (2k-1)\pi \end{cases} \tag{2.12}$$

Where $A_\varphi = \sqrt{\frac{1-j\cot\varphi}{2\pi}}, \varphi = \frac{\pi}{2}\alpha$. The additivity and reversibility properties are its two basic properties which are

$$F^\alpha\left(F^\beta(f(t))\right) = F^\beta\left(F^\alpha(f(t))\right) = F^{\alpha+\beta}(f(t)) \tag{2.13}$$

And

$$F^{-\alpha}\left(F^\alpha(f(t))\right) = f(t) \tag{2.14}$$

When $\varphi = \frac{(2k-1)\pi}{2}, k \in \mathbb{Z}$, FRFT reduces to the classical FT

$$(Ff)(s) = F(f(t))(s) = \sqrt{\frac{1}{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-jst}\mathrm{d}t \tag{2.15}$$

Based on the FRFT and the FT, the definition of fractional cosine transform (FRCT) and fractional sine transform (FRST) have been derived recently. In [19], authors have derived the fractional cosine transform (FRCT) and fractional sine transform (FRST) by taking the real and imaginary parts of the kernel of FRFT

$$F_c^\alpha(f(t))(s) = \int_{-\infty}^{\infty} f(t)\mathrm{Re}[K_F^\varphi(t,s)]\mathrm{d}t,\ s > 0 \tag{2.16}$$

$$F_s^\alpha(f(t))(s) = \int_{-\infty}^{\infty} f(t)\mathrm{Im}[K_F^\varphi(t,s)]\mathrm{d}t,\ s > 0 \tag{2.17}$$

but the fractional cosine transforms (FRCT) and fractional sine transforms (FRST) defined in Eq. (2.16)-(2.17) do not have the additivity property.

In order to overcome the difficulties mentioned above, the authors [20] derived FRCT and FRST by using the derivation process similar to [1] as follows:

$$F_c^\alpha(f(t))(s) = \int_{-\infty}^{\infty} f(t)K^\varphi(t,s)\cos(\csc\varphi \cdot st)\mathrm{d}t,\ s > 0 \tag{2.18}$$

$$F_s^\alpha(f(t))(s) = e^{j\left(\varphi - \frac{\pi}{2}\right)} \int_{-\infty}^{\infty} f(t)K^\varphi(t,s)\cdot\sin(\csc\varphi \cdot st)\mathrm{d}t,\ s > 0 \tag{2.19}$$

The FRCT and the FRST defined as Eq. (2.18) and (2.19) satisfy the additivity property and hence they have the inverse transforms. Moreover, it is better to use the FRCT to process even functions and use the FRST to process odd functions according to [20].

In [21], considering the fact that the FRFT of an even function is even, while the FRFT of an odd function is odd. the authors derived the FRCT and the FRST which are similar to that of Eq. (2.18) and (2.19) as follows

$$F_c^\alpha(f(t))(s) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t)K_\alpha(t,s)\cos(\csc\alpha \cdot st)\mathrm{d}t,\ s > 0 \tag{2.20}$$

$$F_s^\alpha(f(t))(s) = -je^{j\alpha}\sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t)K_\alpha(t,s)\cdot\sin(\csc\alpha \cdot st)\mathrm{d}t,\ s > 0 \tag{2.21}$$

Where $K_\alpha(t,s) = \frac{e^{j\frac{1}{2}\alpha}}{\sqrt{j\sin\alpha}} e^{j\left(\frac{t^2+s^2}{2}\right)\cot\alpha}$.

In this paper, we restrict ourselves to one-sided function $f(x)$, with $f(x) = 0$ for $x < 0$, and use definition for the FRCT and the FRST respectively

$$(F_c^\alpha f)(s) = F_c^\alpha(f(t))(s) = 2\int_0^{\infty} f(t)K^\varphi(t,s)\cos(\csc\varphi \cdot st)\mathrm{d}t,\ s > 0 \tag{2.22}$$

$$(F_s^\alpha f)(s)) = F_s^\alpha(f(t))(s) = 2e^{j\left(\varphi - \frac{\pi}{2}\right)} \int_0^{\infty} f(t)K^\varphi(t,s)\cdot\sin(\csc\varphi \cdot st)\mathrm{d}t,\ s > 0 \tag{2.23}$$

Where $f(x) \in L(\mathbf{R}_+)$, $\varphi = \frac{\pi\alpha}{2}$. The inverse transform of FRCT and FRST are given by

$$f(t) = F_c^{-\alpha}[(F_c^\alpha f)(s)](t) = 2\int_0^{\infty} (F_c^\alpha f)(s)K^{-\varphi}(s,t)\cos(\csc\varphi \cdot ts)\mathrm{d}s,\ t > 0 \tag{2.24}$$

When $f(t)$ is even. And

$$f(t) = F_s^{-\alpha}[(F_s^\alpha f)(s)](t) = -2e^{-j\left(\varphi + \frac{\pi}{2}\right)} \int_0^{\infty} (F_s^\alpha f)(s)K^{-\varphi}(s,t)\cdot\sin(\csc\varphi \cdot ts)\mathrm{d}s,\ t > 0 \tag{2.25}$$

when $f(t)$ is odd.

The definition of FRCT and FRST in [20,21] is different in form, but they are exact same idea. The definition of FRCT and FRST in formula (3.22) and (3.23) are based on [20]. If the definition [21] is used, then the convolution operation $\underset{F_c^\alpha}{\star}$ for the FRCT and the convolution operation $\underset{F_s^\alpha}{\star}$ for the FRST are not influenced as well as the convolution theorem except for the factor differential.

Based on the definition of FRCT and FRST, we can conduct that when $f(t)$ is even,

$$(F_c^\alpha f)(s) = (F^\alpha f)(s)$$

And when $f(t)$ is odd,

$$(F_s^\alpha f)(s) = e^{j\varphi}(F^\alpha f)(s)$$

so the performances of the FRCT, FRST are similar to the performances of the FRFT, and the complexity of computation is one half of the complexity of the FRFT, Therefore, the one-sided FRCT and FRST are more efficient than the FRFT when processing an even and odd signals [20].

When $\varphi = \frac{(2k-1)\pi}{2}$, $k \in \mathbb{Z}$, Eq. (2.22) and (2.23) turns out to be the FCT and the FST in [22]. So, many properties of FCT and FST can be generalized to the FRCT and FRST.

3. The main results

In this section, we will give four novel convolution operations for the FRCT and FRST. The corresponding convolution theorems and product theorems for FRCT and FRST are derived in details. And the relationships among these novel definitions are also investigated.

3.1. The convolution for FRCT and FRST

We first give the definition of convolution operation $\underset{F_c^\alpha,F_s^\alpha}{\star}$ for the FRCT-FRST in Definition 1 and convolution operation for $\underset{F_c^\alpha}{\star}$ the FRCT in Definition 2.

Definition 1

For any two functions $f(t), g(t) \in L(\mathbf{R}_+)$, the convolution operation $\underset{F_c^\alpha,F_s^\alpha}{\star}$ for the FRCT-FRST is defined as follows

$$(f \underset{F_c^\alpha,F_s^\alpha}{\star} g)(t) = A_\varphi e^{-C_{t,\varphi}} \int_0^\infty \tilde{f}(\tau) \left[\tilde{g}(t+\tau) - \text{sign}(t-\tau)\tilde{g}(|t-\tau|)\right] d\tau \tag{3.26}$$

$L(R_+)$ denote the set of all functions $f(t)$ defined on $(0, \infty)$, such that $\int_0^\infty |f(t)|dt < 1$. Where

$$\begin{gathered} A_\varphi = \sqrt{\frac{1-j\cot\varphi}{2\pi}},\ C_{t,\varphi} = j\frac{t^2}{2}\cot\varphi \\ \tilde{f}(t) = f(t)e^{C_{t,\varphi}},\ \tilde{g}(t) = g(t)e^{C_{t,\varphi}}. \end{gathered} \tag{3.27}$$

Definition 2

For any two functions $f(t), g(t) \in L(\mathbf{R}_+)$, the convolution operation $\underset{F_c^\alpha}{\star}$ for the FRCT is defined as follows

$$(f \underset{F_c^\alpha}{\star} g)(t) = A_\varphi e^{-C_{t,\varphi}} \int_0^\infty \tilde{f}(\tau) \left[\tilde{g}(|t-\tau|) + \tilde{g}(t+\tau)\right] d\tau \tag{3.28}$$

Where $A_\varphi, C_{t,\varphi}, \tilde{f}(t), \tilde{g}(t)$ are same to Eqs. (3.27).

Then, we will derive the convolution expressions for the FRCT-FRST. This leads to the following convolution theorem.

Theorem 1

Assume $k_1(t) = (f \underset{F_c^\alpha,F_s^\alpha}{\star} g)(t)$, and $f(t), g(t), k_2(t) \in L(\mathbf{R}_+)$. $F_c^\alpha f, F_c^\alpha k_2, F_s^\alpha g$ denote the FRCT of $f(t), k_2(t)$ and the FRST of $g(t)$ respectively, then we obtain

$$D_\varphi e^{C_{s,\varphi}} F_c^\alpha \left[k_1(t)\right](s) = \left(F_s^\alpha f\right)(s) \left(F_s^\alpha g\right)(s),\ s > 0 \tag{3.29}$$

Where $D_\varphi = e^{j\left(\varphi - \frac{\pi}{2}\right)}, C_{s,\varphi} = j\left(\frac{s^2}{2}\right)\cot\varphi$.

Proof

Based on Eq. (3.26) and the fact that $f(t), g(t) \in L(\mathbf{R}_+)$, we have

$$\begin{aligned} \left(F_s^\alpha f\right)(s) \left(F_s^\alpha g\right)(s) &= B_{s,\varphi} D_\varphi^2 \int_0^\infty \int_0^\infty \sin(\csc\varphi \cdot su)\sin(\csc\varphi \cdot sv) \\ &\quad \cdot e^{j\left(\frac{u^2+v^2}{2}\right)\cot\varphi} f(u)g(v) du dv \\ &= \frac{B_{s,\varphi} D_\varphi^2}{2} \int_0^\infty \int_0^\infty e^{j\left(\frac{u^2+v^2}{2}\right)\cot\varphi} f(u)g(v) \\ &\quad \cdot \left[\cos(s\csc\varphi(u-v)) - \cos(s\csc\varphi(u+v))\right] du dv \\ &= \frac{B_{s,\varphi} D_\varphi^2}{2} \int_0^\infty \int_{-x}^\infty \cos(\csc\varphi \cdot s\omega) f(x) g(\omega+x) e^{j\left(\frac{x^2+(\omega+x)^2}{2}\right)\cot\varphi} dx d\omega \\ &\quad - \frac{B_{s,\varphi} D_\varphi^2}{2} \int_0^\infty \int_x^\infty \cos(\csc\varphi \cdot s\omega) f(x) g(\omega-x) e^{j\left(\frac{x^2+(\omega-x)^2}{2}\right)\cot\varphi} dx d\omega \end{aligned}$$

So one finds

$$\begin{aligned} \left(F_s^\alpha f\right)(s) \left(F_s^\alpha g\right)(s) &= \frac{B_{s,\varphi} D_\varphi^2}{2} \int_0^\infty f(x) \left[\int_0^\infty \cos(\csc\varphi \cdot s\omega) g(\omega+x) e^{j\left(\frac{x^2+(\omega+x)^2}{2}\right)\cot\varphi} d\omega \right. \\ &\quad - \int_x^\infty \cos(\csc\varphi \cdot s\omega) g(\omega-x) e^{j\left(\frac{x^2+(\omega-x)^2}{2}\right)\cot\varphi} d\omega \\ &\quad \left. + \int_0^x \cos(\csc\varphi \cdot s\omega) g(x-\omega) e^{j\left(\frac{x^2+(\omega-x)^2}{2}\right)\cot\varphi} d\omega \right] dx \\ &= \frac{B_{s,\varphi} D_\varphi^2}{2} \int_0^\infty \cos(\csc\varphi \cdot s\omega) \left[\int_0^\infty f(x) \left[g(\omega+x) e^{j\left(\frac{x^2+(\omega+x)^2}{2}\right)\cot\varphi} d\omega \right.\right. \\ &\quad \left.\left. - \text{sign}(\omega-x) g(|\omega-x|) e^{j\left(\frac{x^2+(\omega-x)^2}{2}\right)\cot\varphi} dx \right] d\omega \right. \end{aligned} \tag{3.30}$$

From the Definition 1, we get

$$\begin{aligned}\left(F_s^{\alpha} f\right)(s)\left(F_s^{\alpha} g\right)(s) &= 2D_{\varphi}^2 e^{C_{s,\varphi}} \int_0^{\infty}\left[(f \underset{F_c^{\alpha},F_s^{\alpha}}{\star} g)(\omega)\right] K^{\varphi}(\omega,s)\cos(\csc\varphi\cdot s\omega)\mathrm{d}\omega \\ &= D_{\varphi} e^{C_{s,\varphi}} F_c^{\alpha}\left[k_1(\omega)\right](s)\end{aligned} \tag{3.31}$$

The proof of Theorem 1 is achieved. □

Next, we derive the convolution expressions for the FRCT as follows.

Theorem 2
Assume $k_2(t) = (f \underset{F_c^{\alpha}}{\star} g)(t)$, and $f(t), g(t), k_1(t) \in L(\mathbf{R}_+)$. $F_c^{\alpha}f, F_c^{\alpha}g, F_c^{\alpha}k_1$ denote the FRCT of $f(t)$, $g(t)$ and $k_1(t)$ respectively, then we obtain

$$e^{C_{s,\varphi}} F_c^{\alpha}\left[k_2(t)\right](s) = \left(F_c^{\alpha} f\right)(s)\left(F_c^{\alpha} g\right)(s),\ s>0 \tag{3.32}$$

Where $C_{s,\varphi} = j(\frac{s^2}{2})\cot\varphi$.

Proof
The proof of Theorem 2 is similar to Theorem 1, and it is omitted. □

When $\varphi = \frac{(2k-1)\pi}{2}, k\in\mathbb{Z}$, the convolution theorem for FCT and FST [24] in the Eq. (2.6) and the convolution theorem for FCT [23] in the Eq. (2.4), are all special case of our results in the Eq. (3.29) and (3.32).

3.2. *The convolution with a weight-function for FRCT and FRST*

Based on the Definition 1 and Definition 2, we give the convolution operation with the weight function $\gamma(\tau) = \sin\tau$ of functions f and g for FRST and the weight function $\gamma(\tau) = \cos\tau$ of functions f and g for the FRCT respectively.

Definition 3
For any two functions $f(t), g(t) \in L(\mathbf{R}_+)$, the convolution operation $\underset{F_s^{\alpha}}{\overset{\gamma}{\star}}$ with the weight function $\gamma(\tau) = \sin\tau$ for the FRST is defined as follows

$$\begin{aligned}(f \underset{F_s^{\alpha}}{\overset{\gamma}{\star}} g)(t) = \frac{A_{\varphi}}{2} e^{-C_{t,\varphi}} \int_0^{\infty} & \tilde{f}(\tau)\left[-\tilde{g}(t+\tau+\sin\varphi) + \operatorname{sign}(t-\tau+\sin\varphi)\tilde{g}(|t-\tau+\sin\varphi|)\right. \\ & \left.+\tilde{g}(|t+\tau-\sin\varphi|) - \operatorname{sign}(t-\tau-\sin\varphi)\tilde{g}(|t-\tau-\sin\varphi|)\right]\mathrm{d}\tau\end{aligned} \tag{3.33}$$

Definition 4
For any two functions $f(t), g(t) \in L(\mathbf{R}_+)$, the convolution operation $\underset{F_c^{\alpha}}{\overset{\gamma}{\star}}$ with the weight function $\gamma(\tau) = \cos\tau$ for the FRCT is defined as follows

$$\begin{aligned}(f \underset{F_c^{\alpha}}{\overset{\gamma}{\star}} g)(t) = \frac{A_{\varphi}}{2} e^{-C_{t,\varphi}} \int_0^{\infty} & \tilde{f}(\tau)\cdot \\ & \left[\tilde{g}(|t-\tau-\sin\varphi|) + \tilde{g}(t+\tau+\sin\varphi) + \tilde{g}(|t-\tau+\sin\varphi|) + \tilde{g}(|t+\tau-\sin\varphi|)\right]\mathrm{d}\tau\end{aligned} \tag{3.34}$$

Where $A_{\varphi}, C_{t,\varphi}, \tilde{f}(t), \tilde{g}(t)$ are same to Eqs. (3.27).

And then, we obtain the convolution theorem with a weight function $\gamma(\tau) = \sin\tau$ for the FRST based on the Definition 3.

Theorem 3
Assume $k_3(t) = (f \underset{F_s^{\alpha}}{\overset{\gamma}{\star}} g)(t)$, and $f(t), g(t), k_3(t) \in L(\mathbf{R}_+)$. $F_s^{\alpha}f, F_s^{\alpha}g, F_s^{\alpha}k_3$ denote the FRST of $f(t)$, $g(t)$ and $k_3(t)$, then the convolution theorem with a weight function $\gamma(\tau) = \sin\tau$ for FRST is obtained:

$$e^{C_{s,\varphi}} e^{j(\varphi-\frac{\pi}{2})} F_s^{\alpha}\left[k_3(t)\right](s) = \sin s\left(F_s^{\alpha} f\right)(s)\left(F_s^{\alpha} g\right)(s),\quad \forall s>0 \tag{3.35}$$

Proof
Based on Eq. (3.39) and the fact that $f(t), g(t) \in L(\mathbf{R}_+)$, we have

$$\begin{aligned}\sin s\left(F_s^{\alpha} f\right)(s)\left(F_s^{\alpha} g\right)(s) &= B_{s,\varphi} D_{\varphi}^2 \int_0^{\infty}\int_0^{\infty} \sin s\sin(\csc\varphi\cdot su)\sin(\csc\varphi\cdot sv) \\ &\qquad \cdot f(u)g(v)e^{j(\frac{u^2+v^2}{2})\cot\varphi}\mathrm{d}u\mathrm{d}v \\ &= \frac{B_{s,\varphi} D_{\varphi}^2}{4} \int_0^{\infty}\int_0^{\infty} \left[\sin s(\csc\varphi\cdot(u+v)-1)\right. \\ &\qquad + \sin s(\csc\varphi\cdot(u-v)+1) - \sin s(\csc\varphi\cdot(u+v)+1) \\ &\qquad \left.- \sin s(\csc\varphi\cdot(u-v)-1)\right]\cdot f(u)g(v)e^{j\left(\frac{u^2+v^2}{2}\right)\cot\varphi}\mathrm{d}u\mathrm{d}v\end{aligned} \tag{3.36}$$

And

$$\begin{aligned}
&\int_0^\infty\int_0^\infty [\sin s(\csc\varphi\cdot(u-v)+1)-\sin s(\csc\varphi\cdot(u+v)+1)]\cdot e^{j\left(\frac{u^2+v^2}{2}\right)\cot\varphi}f(u)g(v)\mathrm{d}u\mathrm{d}v\\
&=-\int_0^\infty\int_0^\infty \sin(\csc\varphi\cdot s\omega)f(x)g(\omega+x+\sin\varphi)e^{j\left(\frac{x^2+(\omega+x+\sin\varphi)^2}{2}\right)\cot\varphi}\mathrm{d}x\mathrm{d}\omega\\
&\quad+\int_0^\infty\int_0^{x+\sin\varphi} \sin(\csc\varphi\cdot s\omega)f(x)g(x-\omega+\sin\varphi)e^{j\left(\frac{x^2+(x-\omega+\sin\varphi)^2}{2}\right)\cot\varphi}\mathrm{d}x\mathrm{d}\omega\\
&\quad-\int_0^\infty\int_{x+\sin\varphi}^\infty \sin(\csc\varphi\cdot s\omega)f(x)g(\omega-x-\sin\varphi)e^{j\left(\frac{x^2+(\omega-x-\sin\varphi)^2}{2}\right)\cot\varphi}\mathrm{d}x\mathrm{d}\omega\\
&=\int_0^\infty \sin(\csc\varphi\cdot s\omega)\left[\int_0^\infty \tilde{f}(x)\left[-\tilde{g}(\omega+x+\sin\varphi)\right.\right.\\
&\quad\left.\left.+\mathrm{sign}(x+\sin\varphi-\omega)\tilde{g}(|x+\sin\varphi-\omega|)e^{j\left(\frac{\omega^2}{2}\right)\cot\varphi}\right]\mathrm{d}x\right]\mathrm{d}\omega
\end{aligned}\tag{3.37}$$

While

$$\begin{aligned}
&\int_0^\infty\int_0^\infty [\sin s(\csc\varphi\cdot(u+v)-1)-\sin s(\csc\varphi\cdot(u-v)-1)]\cdot e^{j\left(\frac{u^2+v^2}{2}\right)\cot\varphi}f(u)g(v)\mathrm{d}u\mathrm{d}v\\
&=\int_0^\infty \sin(\csc\varphi\cdot s\omega)\left[\int_0^\infty \tilde{f}(x)\left[\tilde{g}(|x+\omega-\sin\varphi|)\right.\right.\\
&\quad\left.\left.+\mathrm{sign}(\omega-x+\sin\varphi)\tilde{g}(|\omega-x+\sin\varphi|)e^{j\left(\frac{\omega^2}{2}\right)\cot\varphi}\right]\mathrm{d}x\right]\mathrm{d}\omega
\end{aligned}\tag{3.38}$$

So with Eq. (3.42), (3.43), (3.44), we obtain

$$\begin{aligned}
\sin s\left(F_s^\alpha f\right)(s)\left(F_s^\alpha g\right)(s) &= B_{s,\varphi}D_\varphi^2\int_0^\infty\int_0^\infty \sin(\csc\varphi\cdot s\omega)\\
&\left[\tilde{f}(x)\left[-\tilde{g}(\omega+x+\sin\varphi)+\mathrm{sign}(\omega-x+\sin\varphi)\tilde{g}(|\omega-x+\sin\varphi|)\right.\right.\\
&\left.\left.+\tilde{g}(|\omega+x-\sin\varphi|)+\mathrm{sign}(\omega-x-\sin\varphi)\tilde{g}(|\omega-x-\sin\varphi|)\right]\right]\mathrm{d}x\mathrm{d}\omega
\end{aligned}\tag{3.39}$$

From the Definition 3, we obtain

$$\begin{aligned}
&\sin s\left(F_s^\alpha f\right)(s)\left(F_s^\alpha g\right)(s)\\
&=2e^{C_{s,\varphi}}D_\varphi\int_0^\infty\left(f(x)\underset{F_s^\alpha}{\overset{\gamma}{\star}}g(x)\right)K^\varphi(\omega,s)\sin(\csc\varphi\cdot s\omega)\mathrm{d}\omega\\
&=e^{C_{s,\varphi}}D_\varphi F_s^\alpha\left[k_3(\omega)\right](s)
\end{aligned}\tag{3.40}$$

The proof of Theorem 3 is achieved. □

Similarly, we obtain the convolution theorem with a weight function $\gamma(\tau)=\cos\tau$ for the FRCT based on the Definition 4.

Theorem 4

Assume $k_4(t)=(f\underset{F_c^\alpha}{\overset{\gamma}{\star}}g)(t)$, and $f(t), g(t), k_4(t)\in L(\mathbf{R}_+)$. $F_c^\alpha f, F_c^\alpha g, F_c^\alpha k_4$ denote the FRCT of $f(t)$, $g(t)$ and $k_4(t)$, then the convolution theorem with a weight function $\gamma(\tau)=\cos\tau$ for FRCT is obtained:

$$e^{C_{s,\varphi}}F_c^\alpha\left[k_4(t)\right](s)=\cos s\left(F_c^\alpha f\right)(s)\left(F_c^\alpha g\right)(s),\quad \forall s>0\tag{3.41}$$

Proof

The proof of Theorem 4 is similar to Theorem 3, and it is omitted. □

Based on the Definition 3 and the Theorem 3, using the convolution operation $(f\underset{F_s^\alpha}{\overset{\gamma}{\star}}g)(t)$ with the weight function $\gamma(\tau)=\sin\tau$, we obtain the convolution theorem for the FRST in Eq. (3.35). It shows that the FRST of convolution of two functions is equivalent to simple multiplication of their FRST domain. Similarly, based on the Definition 4 and the Theorem 4, using the convolution operation$(f\underset{F_c^\alpha}{\overset{\gamma}{\star}}g)(t)$ with the weight function $\gamma(\tau)=\cos\tau$, the convolution theorem for the FRCT in Eq. (3.41) is obtained. It suggests that the FRCT of convolution of two functions is equivalent to simple multiplication of their FRCT domain. It should be pointed out that only using the convolution operation $(f\underset{F_s^\alpha}{\overset{\gamma}{\star}}g)(t)$ with the weight function $\gamma(\tau)=\sin\tau$, and the convolution operation$(f\underset{F_c^\alpha}{\overset{\gamma}{\star}}g)(t)$ with the weight function $\gamma(\tau)=\cos\tau$, can we obtain this very nice and simply convolution structure which is similar to [5–11]. This may be particularly useful in filter design and applications as will be shown later on.

Next, we will derive the convolution theorem for FRCT and FRST of the pruduct of two functions. This leads to the following convolution theorems

Theorem 5
Let $f(t), g(t) \in L_1(\mathbf{R}_+)$. $k_5(t) = f(t) \cdot g(t)$, $F_c^\alpha(\cdot)$, $F_s^\alpha(\cdot)$ denote the FRCT operation and FRST operation, then the convolution theorem for FRCT and FRST of the pruduct of two functions $k_5(t)$ is obtained

$$F_c^\alpha[k_5(t)](s) = 2\left[1+\cot^2(\varphi)\right] e^{C_{s,\varphi}} \left[\left((F_c^\alpha g)(s)e^{-C_{s,\varphi}}\right) \underset{F_c}{\star} (F_c f)(s \cdot \csc\varphi)\right] \tag{3.42}$$

And

$$F_s^\alpha[k_5(t)](s) = -2\left[1+\cot^2(\varphi)\right] e^{-j\pi} e^{C_{s,\varphi}} \left[\left((F_s^\alpha g)(s)e^{-C_{s,\varphi}}\right) \underset{1}{\star} (F_c f)(s \cdot \csc\varphi)\right] \tag{3.43}$$

respectively, where $\underset{F_c}{\star}$ denotes the convolution for FCT in Eq. (2.3), and $\underset{1}{\star}$ denotes the convolution for FST-FCT in Table I.

Proof
Based on Eq. (2.22), (2.24) and the fact that $f(t), g(t) \in L_1(\mathbf{R}_+)$, we have

$$\begin{aligned}
F_c^\alpha[k_5(t)](s) &= F_c^\alpha[f(t)g(t)](s) = 2\int_0^\infty f(t)g(t)K^\varphi(t,s)\cos(\csc\varphi \cdot st)\mathrm{d}t \\
&= 4\int_0^\infty f(t)K^\varphi(t,s)\cos(\csc\varphi \cdot st)\left[\int_0^\infty (F_c^\alpha g)(\omega)K^{-\varphi}(\omega,t)\cos(\csc\varphi \cdot t\omega)\mathrm{d}\omega\right]\mathrm{d}t \\
&= \frac{1+\cot^2\varphi}{\pi} e^{C_{s,\varphi}} \int_0^\infty (F_c^\alpha g)(\omega)e^{-C_{\omega,\varphi}} \\
&\qquad \cdot \left[\int_0^\infty f(t)\cos(\csc\varphi \cdot t(s+\omega)) + \cos(\csc\varphi \cdot t(s-\omega)\mathrm{d}t\right]\mathrm{d}\omega \\
&= \sqrt{\frac{2}{\pi}}(1+\cot^2\varphi)e^{C_{s,\varphi}} \int_0^\infty (F_c^\alpha g)(\omega)e^{-C_{\omega,\varphi}} \\
&\qquad \cdot \left[(F_c f)(\csc\varphi(s+\omega)) + (F_c f)(\csc\varphi|s-\omega|)\right]\mathrm{d}\omega \\
&= 2\left[1+\cot^2(\varphi)\right] e^{C_{s,\varphi}} \left[\left((F_c^\alpha g)(s)e^{-C_{s,\varphi}}\right) \underset{F_c}{\star} (F_c f)(s \cdot \csc\varphi)\right]
\end{aligned} \tag{3.44}$$

This is completing the proof of Eq. (3.42), the proof for the Eq. (3.43) is similar to that of the Eq. (3.42) based on the Eq. (2.23), (2.25). □

Corollary 1
When $\varphi = \frac{(2k-1)\pi}{2}, k \in \mathbb{Z}$, then the convolution Theorem 1 and Theorem 2 reduces to the convolution in FCT-FST domain [24] and convolution in FCT domain [23] as follows:

$$f(t) \underset{F_c}{\star} g(t) \overset{\text{FCT}}{\longleftrightarrow} (F_c f)(s)(F_c g)(s) \tag{3.45}$$

And

$$f(t) \underset{2}{\star} g(t) \overset{\text{FCT}}{\longleftrightarrow} (F_s f)(s)(F_s g)(s) \tag{3.46}$$

respectively, where $\underset{F_c}{\star}$ denotes the convolution for FCT in Eq. (2.3), and $\underset{2}{\star}$ denotes the convolution for FCT-FST in Eq. (2.5).

Corollary 2
When $\varphi = \frac{(2k-1)\pi}{2}, k \in \mathbb{Z}$, then the convolution Theorem 3 and Theorem 4 reduces to the convolution in FCT and FST domain as follows:

$$f(t) \overset{\gamma}{\underset{F_s}{\star}} g(t) \overset{\text{FST}}{\longleftrightarrow} \sin s\,(F_s f)(s)(F_s g)(s) \tag{3.47}$$

And

$$f(t) \overset{\gamma}{\underset{F_c}{\star}} g(t) \overset{\text{FCT}}{\longleftrightarrow} \cos s\,(F_c f)(s)(F_c g)(s) \tag{3.48}$$

respectively, where $\overset{\gamma}{\underset{F_s}{\star}}$ denotes the convolution for FST in Eq. (2.7), and $\overset{\gamma}{\underset{F_c}{\star}}$ denotes the convolution for FCT in Eq. (2.9).

3.3. *The relationships between the different definition of convolution operations*

In this section, we will discuss the relationships of Definition 1-4. When $\varphi = \frac{(2k-1)\pi}{2}, k \in \mathbb{Z}$, then the FRCT and FRST reduces to the FCT and FST respectively, hence Eq. (2.3), (2.5), (2.7) and (2.9) are all special case of our result which is in Eq. (3.26), (3.28), (3.33) and (3.34). Meanwhile, the convolution operation in Eq. (3.26), (3.28) can be expressed as the known convolution in Eq. (2.3), (2.5) as follows.

Theorem 6

Let $f(t), g(t) \in L_1(\mathbf{R}_+)$. $F_c^\alpha(\cdot)$, $F_s^\alpha(\cdot)$ denote the FRCT operation and FRST operation, then the convolution operation $\underset{F_c^\alpha,F_s^\alpha}{\star}$ for FRCT-FRST and $\underset{F_c^\alpha}{\star}$ for FRCT can be rewrited respectively

$$(f \underset{F_c^\alpha,F_s^\alpha}{\star} g)(t) = \sqrt{1 - j\cot\varphi}\, e^{-C_{t,\varphi}} \left(\tilde{f}(t) \underset{2}{\star} \tilde{g}(t)\right) \tag{3.49}$$

$$(f \underset{F_c^\alpha}{\star} g)(t) = \sqrt{1 - j\cot\varphi}\, e^{-C_{t,\varphi}} \left(\tilde{f}(t) \underset{F_c}{\star} \tilde{g}(t)\right) \tag{3.50}$$

where $\tilde{f}(t) = f(t)e^{C_{t,\varphi}}$, $\tilde{g}(t) = g(t)e^{C_{t,\varphi}}$, $\underset{F_c}{\star}$ is the convolution operation for the FCT in Eq. (2.3), and $\underset{2}{\star}$ is the convolution operation for the FCT-FST in Eq. (2.5).

Proof

The proof follows easily from Definition 1-2 and Eq. (2.3), (2.5), hence it is omitted. □

See Figure 1 and Figure 2 for a realization of the convolution operation $\underset{F_c^\alpha,F_s^\alpha}{\star}$ and $\underset{F_c^\alpha}{\star}$. Next, we will discuss the relation between the convolution operation $\overset{\gamma}{\underset{F_c^\alpha}{\star}}$, $\overset{\gamma}{\underset{F_s^\alpha}{\star}}$ and the convolution operation $\underset{F_c}{\star}$, $\underset{2}{\star}$. That is, the convolution operation in Eq. (3.33) and Eq. (3.34) can be expressed as another form as follows.

Theorem 7

Let $f(t), g(t) \in L_1(\mathbf{R}_+)$. $F_c^\alpha(\cdot)$, $F_s^\alpha(\cdot)$ denote the FRCT operation and FRST operation, then the convolution operation $\overset{\gamma}{\underset{F_c^\alpha}{\star}}$ for FRCT and $\overset{\gamma}{\underset{F_s^\alpha}{\star}}$

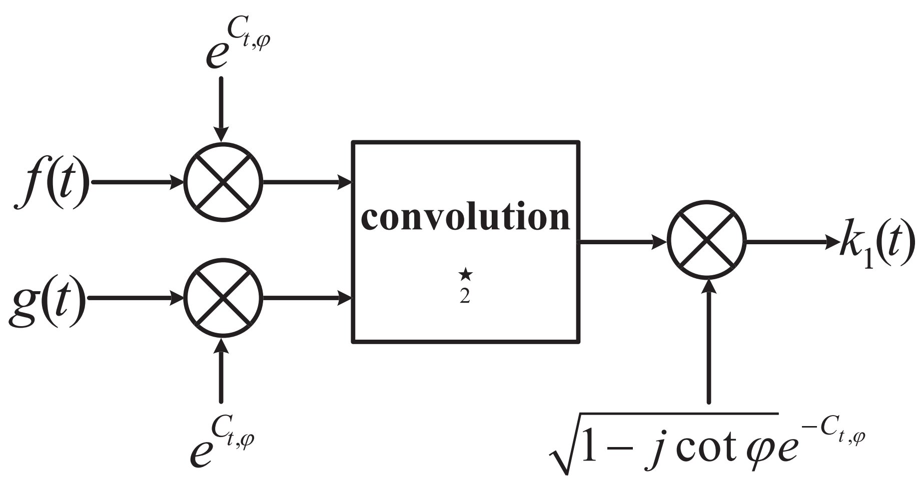

Figure 1. The convolution operation $\underset{F_c^\alpha,F_s^\alpha}{\star}$ for the FRCT-FRST.

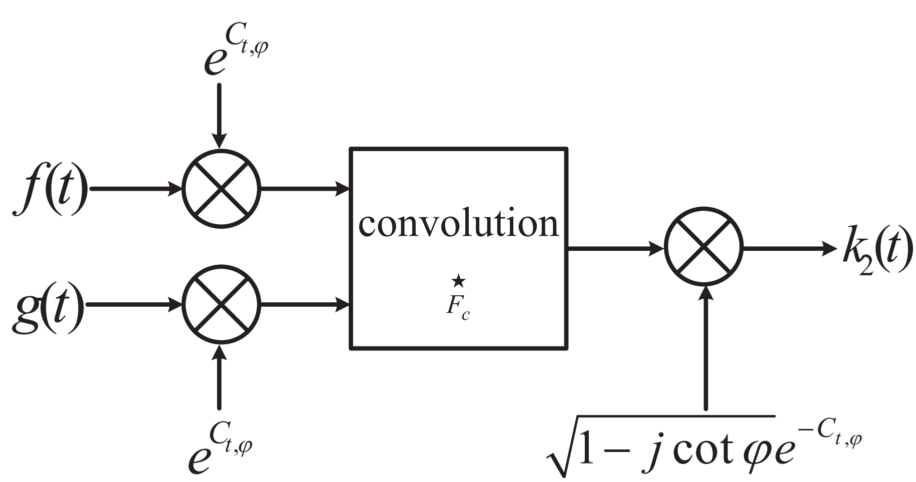

Figure 2. The convolution operation $\underset{F_c^\alpha}{\star}$ for the FRCT.

for FRST can be rewrited respectively.

$$(f \overset{\gamma}{\underset{F_s^\alpha}{\star}} g)(t) = \frac{1-j\cot\varphi}{2\sqrt{2\pi}} e^{-2C_{t,\varphi}} \left[\left(\bar{f}(t) \underset{2}{\star} \bar{g}(t) \right) (|t-\sin\varphi|) - \left(\bar{f}(t) \underset{2}{\star} \bar{g}(t) \right) (t+\sin\varphi) \right] \tag{3.51}$$

And

$$(f \overset{\gamma}{\underset{F_c^\alpha}{\star}} g)(t) = \frac{1-j\cot\varphi}{2\sqrt{2\pi}} e^{-2C_{t,\varphi}} \left[\left(\bar{f}(t) \underset{F_c}{\star} \bar{g}(t) \right) (|t-\sin\varphi|) + \left(\bar{f}(t) \underset{F_c}{\star} \bar{g}(t) \right) (t+\sin\varphi) \right] \tag{3.52}$$

where $\bar{f}(t) = f(t)e^{2C_{t,\varphi}}$ and $\bar{g}(t) = g(t)e^{2C_{t,\varphi}}$.

Proof
We first proof Eq. (3.51). Based on Eq. (3.33) and the fact that $f(t), g(t) \in L_1(\mathbf{R}_+)$, when $t > \sin\varphi > 0$, we have

$$\begin{aligned}
(f \overset{\gamma}{\underset{F_s^\alpha}{\star}} g)(t) &= -\frac{1}{2}e^{-C_{t,\varphi}} \left[A_\varphi \int_0^\infty \tilde{f}(\tau) \left[\tilde{g}(t+\sin\varphi+\tau) - \operatorname{sign}(t+\sin\varphi-\tau)\tilde{g}(|t+\sin\varphi-\tau|) \right] \mathrm{d}\tau \right] \\
&\quad + \frac{1}{2}e^{-C_{t,\varphi}} \left[A_\varphi \int_0^\infty \tilde{f}(\tau) \left[\tilde{g}(t-\sin\varphi+\tau) - \operatorname{sign}(t-\sin\varphi-\tau)\tilde{g}(|t-\sin\varphi-\tau|) \right] \mathrm{d}\tau \right] \\
&= \frac{1}{2}e^{-C_{t,\varphi}} \left[(f \underset{F_c^\alpha, F_s^\alpha}{\star} g)(t-\sin\varphi) - (f \underset{F_c^\alpha, F_s^\alpha}{\star} g)(t+\sin\varphi) \right]
\end{aligned} \tag{3.53}$$

Similarly, when $0 < t < \sin\varphi$, we have

$$\begin{aligned}
(f \overset{\gamma}{\underset{F_s^\alpha}{\star}} g)(t) &= -\frac{1}{2}e^{-C_{t,\varphi}} \left[A_\varphi \int_0^\infty \tilde{f}(\tau) \left[\tilde{g}(t+\sin\varphi+\tau) - \operatorname{sign}(t+\sin\varphi-\tau)\tilde{g}(|t+\sin\varphi-\tau|) \right] \mathrm{d}\tau \right] \\
&\quad + \frac{1}{2}e^{-C_{t,\varphi}} \left[A_\varphi \int_0^\infty \tilde{f}(\tau) \left[\tilde{g}(\sin\varphi-t+\tau) - \operatorname{sign}(\sin\varphi-t-\tau)\tilde{g}(|\sin\varphi-t-\tau|) \right] \mathrm{d}\tau \right] \\
&= \frac{1}{2}e^{-C_{t,\varphi}} \left[(f \underset{F_c^\alpha, F_s^\alpha}{\star} g)(\sin\varphi-t) - (f \underset{F_c^\alpha, F_s^\alpha}{\star} g)(t+\sin\varphi) \right]
\end{aligned} \tag{3.54}$$

From Eq. (3.53) and (3.54), we have

$$(f \overset{\gamma}{\underset{F_s^\alpha}{\star}} g)(t) = \frac{1}{2}e^{-C_{t,\varphi}} \left[(f \underset{F_c^\alpha, F_s^\alpha}{\star} g)(|t-\sin\varphi|) - (f \underset{F_c^\alpha, F_s^\alpha}{\star} g)(t+\sin\varphi) \right] \tag{3.55}$$

The Eq. (3.55) and (3.49) yield

$$\begin{aligned}
(f \overset{\gamma}{\underset{F_s^\alpha}{\star}} g)(t) = \frac{1-j\cot\varphi}{2\sqrt{2\pi}} e^{-2C_{t,\varphi}} \Big[& (f(|t-\sin\varphi|)e^{2C_{t-\sin\varphi,\varphi}}) \underset{2}{\star} (g(|t-\sin\varphi|)e^{2C_{t-\sin\varphi,\varphi}}) \\
& -(f(t+\sin\varphi)e^{2C_{t+\sin\varphi,\varphi}}) \underset{2}{\star} (g(t+\sin\varphi)e^{2C_{t+\sin\varphi,\varphi}}) \Big]
\end{aligned}$$

That is

$$(f \overset{\gamma}{\underset{F_s^\alpha}{\star}} g)(t) = \frac{1-j\cot\varphi}{2\sqrt{2\pi}} e^{-2C_{t,\varphi}} \left[\left(\bar{f}(t) \underset{2}{\star} \bar{g}(t) \right) (|t-\sin\varphi|) - \left(\bar{f}(t) \underset{2}{\star} \bar{g}(t) \right) (t+\sin\varphi) \right] \tag{3.56}$$

The proof for the Eq. (3.52) is similar to the Eq. (3.51), and it is omitted. The theorem is proved. □

See Figure 3 and Figure 4 for a realization of the convolution operation $\overset{\gamma}{\underset{F_s^\alpha}{\star}}$ for FRST and $\overset{\gamma}{\underset{F_c^\alpha}{\star}}$ for FRCT.

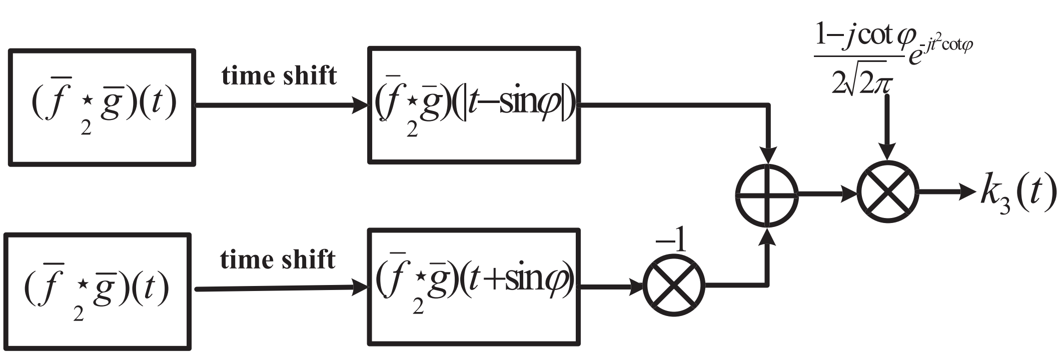

Figure 3. The convolution operation $\overset{\gamma}{\underset{F_s^\alpha}{\star}}$ for the FRST.

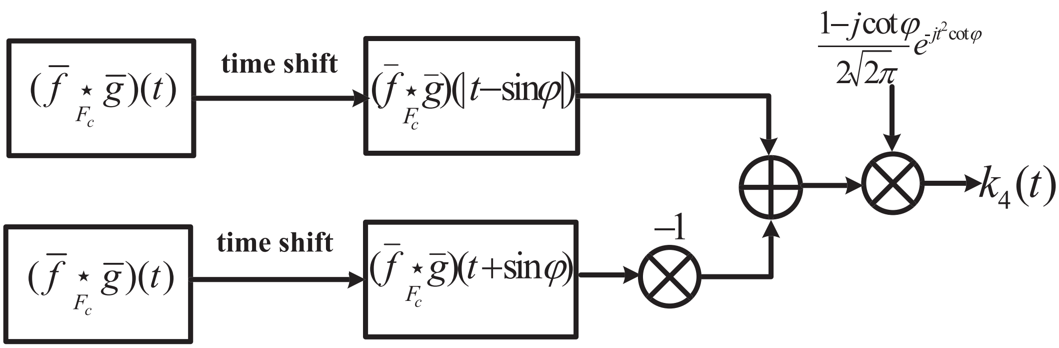

Figure 4. The convolution operation $\overset{\gamma}{\underset{F_c^\alpha}{\star}}$ for the FRCT.

Remark 1

In the space $L_1(\mathbf{R}_+)$, the convolution $\underset{F_c^\alpha,F_s^\alpha}{\star}$, $\underset{F_c^\alpha}{\star}$, $\overset{\gamma}{\underset{F_s^\alpha}{\star}}$ and $\overset{\gamma}{\underset{F_c^\alpha}{\star}}$ in Eq. (3.26), (3.28), (3.33), (3.34) are all commutative, that is

$$f \underset{F_c^\alpha}{\star} g = g \underset{F_c^\alpha}{\star} f,\ f \underset{F_c^\alpha,F_s^\alpha}{\star} g = g \underset{F_c^\alpha,F_s^\alpha}{\star} f,\ f \overset{\gamma}{\underset{F_s^\alpha}{\star}} g = g \overset{\gamma}{\underset{F_s^\alpha}{\star}} f,\ f \overset{\gamma}{\underset{F_c^\alpha}{\star}} g = g \overset{\gamma}{\underset{F_c^\alpha}{\star}} f \tag{3.57}$$

Remark 2

In the space $L_1(\mathbf{R}_+)$, the convolution $\underset{F_c^\alpha}{\star}$, $\overset{\gamma}{\underset{F_s^\alpha}{\star}}$ and $\overset{\gamma}{\underset{F_c^\alpha}{\star}}$ in Eq. (3.28), (3.33), (3.34) are associative, while the convolution $\underset{F_c^\alpha,F_s^\alpha}{\star}$ in Eq. (3.26) is not associative, and they are satisfied the following equalities

$$(1)\ (f \underset{F_c^\alpha}{\star} g) \underset{F_c^\alpha}{\star} h = (f \underset{F_c^\alpha}{\star} h) \underset{F_c^\alpha}{\star} g = (h \underset{F_c^\alpha}{\star} g) \underset{F_c^\alpha}{\star} f = f \underset{F_c^\alpha}{\star} (h \underset{F_c^\alpha}{\star} g) = g \underset{F_c^\alpha}{\star} (f \underset{F_c^\alpha}{\star} h) = h \underset{F_c^\alpha}{\star} (f \underset{F_c^\alpha}{\star} g)$$

$$(2)\ f \overset{\gamma}{\underset{F_c^\alpha}{\star}} (g \overset{\gamma}{\underset{F_c^\alpha}{\star}} h) = g \overset{\gamma}{\underset{F_c^\alpha}{\star}} (f \overset{\gamma}{\underset{F_c^\alpha}{\star}} h) = h \overset{\gamma}{\underset{F_c^\alpha}{\star}} (f \overset{\gamma}{\underset{F_c^\alpha}{\star}} g)$$

$$(3)\ f \overset{\gamma}{\underset{F_s^\alpha}{\star}} (g \overset{\gamma}{\underset{F_s^\alpha}{\star}} h) = g \overset{\gamma}{\underset{F_s^\alpha}{\star}} (f \overset{\gamma}{\underset{F_s^\alpha}{\star}} h) = h \overset{\gamma}{\underset{F_s^\alpha}{\star}} (f \overset{\gamma}{\underset{F_s^\alpha}{\star}} g)$$

$$(4)\ f \overset{\gamma}{\underset{F_c^\alpha}{\star}} (g \underset{F_c^\alpha}{\star} h) = g \overset{\gamma}{\underset{F_c^\alpha}{\star}} (f \underset{F_c^\alpha}{\star} h) = h \overset{\gamma}{\underset{F_c^\alpha}{\star}} (f \underset{F_c^\alpha}{\star} g)$$

$$(5)\ f \underset{F_c^\alpha,F_s^\alpha}{\star} (g \overset{\gamma}{\underset{F_s^\alpha}{\star}} h) = g \underset{F_c^\alpha,F_s^\alpha}{\star} (f \overset{\gamma}{\underset{F_s^\alpha}{\star}} h) = h \underset{F_c^\alpha,F_s^\alpha}{\star} (f \overset{\gamma}{\underset{F_s^\alpha}{\star}} g)$$

When $\varphi = \frac{(2k-1)\pi}{2}, k \in \mathbb{Z}^+$, then, Eq. (3.52) reduce to the Theorem 4 [27]. Due to Theorem 7, Remark 1 and Remark 2, the convolution operation $\overset{\gamma}{\underset{F_c^\alpha}{\star}}$ for FRCT and $\overset{\gamma}{\underset{F_s^\alpha}{\star}}$ for FRST exhibits the commutative and associative properties, therefore, they are particularly powerful for the designing of multiplicative filters in the FRCT and FRST domain through both the new convolution in the time domain and product in the FRCT and FRST domain.

4. Multiplicative filtering for FRCT and FRST

In this section, we mainly use the convolution operation and convolution theorem for the FRCT and FRST to investigate the application in designing of the multiplicative filters. From what has been discussed above, we know that it is convenient to use the FRCT or FRST instead of the FRFT. Hence, we can use the FRCT to process even signals and use the FRST to process odd signals. And that the complexity of the FRCT or FRST is just about one half of the complexity of the FRFT. For this point of view, using the FRCT or FRST for filter design is more efficient than using the FRFT.

4.1. Designing of multiplicative filters through the product in the FRCT and FRST domain

In this subsection, the new weighted convolution theorem is carried out to design multiplicative filters in the FRCT and FRST domain in terms of the product in the FRCT and FRST domain.

In recent years, papers [31–37] discuss the usage of filter designed by FRFT, linear canonical transform (LCT), offset FRFT and offset LCT, to remove distortion or noise. Here, the model of the multiplicative filter in the FRCT or FRST domain is shown in Figure 5. The effect of the multiplicative filter in the FRCT or FRST domain in Figure 5 can be written in the following equation:

$$f_{\text{out}}(t) = F_c^{-\alpha}\left[F^\alpha_{c,f_{\text{in}}}(s) \cdot \widetilde{H}_1(s)\right](t) \tag{4.58}$$

When $f_{\text{in}}(t)$ is even signal. And

$$f_{\text{out}}(t) = F_s^{-\alpha}\left[F^\alpha_{s,f_{\text{in}}}(s) \cdot \widetilde{H}_2(s)\right](t) \tag{4.59}$$

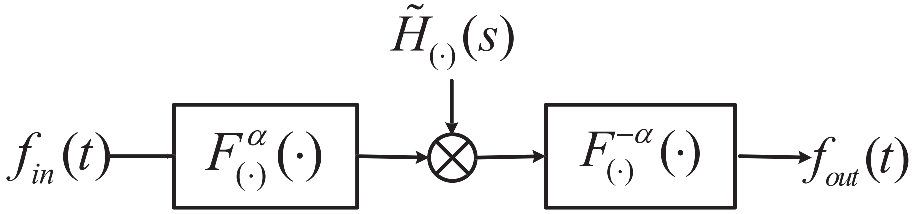

Figure 5. The multiplicative filter in the FRCT or FRST domain.

when $f_{\text{in}}(t)$ is odd signal, where $F^{\alpha}_{c,f_{\text{in}}}(s)$ and $F^{\alpha}_{s,f_{\text{in}}}(s)$ denotes the FRCT and FRST of received signal $f_{\text{in}}(t)$ respectively. When the input of the system is a mixed signal (even signal and odd signal), we can first decompose this signal into the sum of an even signal and an odd signal.

Many models of multiplicative filters can be achieved by designing $\tilde{H}(s)$, such as lowpass, bandpass, and bandstop, and so on. In practical application, if $F_c^{\alpha}(s)$, $F_s^{\alpha}(s)$ be the FRCT, FRST components of the desired signal, and $N_c^{\alpha}(s)$, $N_s^{\alpha}(s)$ be the FRCT, FRST components of the noise respectively, having no overlapping or minimal overlapping. Then, the desired signal can be reserved and the noise can be discarded to a large extent through a multiplicative filter in the FRCT and FRST domain. For example, we are interested only in the frequency spectrum of the FRCT or FRST in the region $[s_1, s_2]$ of the designed signal $f(t)$. According to Theorem 3-4, we choose function

$$\widetilde{H}_1(s) = e^{-j\left(\frac{s^2}{2}\right)\cot\varphi}\cos s\left(F_c^{\alpha}g_1\right)(s),\ \ \widetilde{H}_2(s) = e^{-j\left(\frac{s^2}{2}\right)\cot\varphi}e^{-j(\varphi-\frac{\pi}{2})}\sin s\left(F_s^{\alpha}g_2\right)(s) \tag{4.60}$$

as the transfer function of the multiplicative filter, such that $\widetilde{H}_1(s), \widetilde{H}_2(s)$ is constant over $[s_1, s_2]$ and zero or of rapid delay outside that region. By reversibility property of FRCT or FRST, the designed signal $f(t)$ can be obtained. Which is in accordance with that introduced in [32,35,37]. Therefore, it is easier to implement but is more suited for processing even and odd signals than the methods in [32,35,37].

4.2. *Designing of multiplicative filters through the product in the time domain*

In this subsection, the convolution Theorem 3-4 is performed to design multiplicative filters in the FRCT and FRST domain with respect to the new convolution in the time domain. From Figure 6, the output of the filter is expressed as

$$f_{out}(t) = \frac{1-j\cot\varphi}{2\sqrt{2\pi}}e^{-jt^2\cot\varphi}\left[\left(\bar{f}_{in}(t)\underset{F_c}{\star}h_1(t)\right)(|t-\sin\varphi|) + \left(\bar{f}_{in}(t)\underset{F_c}{\star}h_1(t)\right)(t+\sin\varphi)\right] \tag{4.61}$$

When $f_{in}(t)$ is even signal, where $\underset{F_c}{\star}$ denotes convolution of the FCT in Eq. (2.3). From Figure 7, the output of the filter is expressed as

$$(f_{out}(t) = \frac{1-j\cot\varphi}{2\sqrt{2\pi}}e^{-jt^2\cot\varphi}\left[\left(\bar{f}_{in}(t)\underset{2}{\star}h_2(t)\right)(|t-\sin\varphi|) - \left(\bar{f}_{in}(t)\underset{2}{\star}h_2(t)\right)(t+\sin\varphi)\right] \tag{4.62}$$

When $f_{in}(t)$ is odd signal, where $\underset{2}{\star}$ denotes convolution of the FCT-FST in Eq. (2.5). We can designed convoution function

$$h_1(t) = \frac{1}{2}(1+j\cot\varphi)^{-\frac{1}{2}}e^{j\left(\frac{t^2}{2}\right)\cot\varphi}g_1(t),$$

$$h_2(t) = \frac{1}{2}(1+j\cot\varphi)^{-\frac{1}{2}}e^{j\left(\frac{t^2}{2}\right)\cot\varphi}e^{j\pi}g_2(t)$$

And

$$\begin{aligned}
g_1(t) &= F_c^{-\alpha}\left[\left(F_c^{\alpha}g\right)(\omega)\right](t)\\
&= \sqrt{\frac{2(1+j\cot\varphi)}{\pi}}e^{-j\left(\frac{t^2}{2}\right)\cot\varphi}\int_0^{\infty}\left(F_c^{\alpha}g\right)(\omega)e^{-j\left(\frac{\omega^2}{2}\right)\cot\varphi}\cos(\csc\varphi\cdot t\omega)\mathrm{d}\omega\\
&= \sqrt{\frac{2(1+j\cot\varphi)}{\pi}}e^{-j\left(\frac{t^2}{2}\right)\cot\varphi}\int_0^{\infty}\tilde{H}_1(\omega)\sec\omega\cos(\csc\varphi\cdot t\omega)\mathrm{d}\omega\\
&= 2\sqrt{1+j\cot\varphi}e^{-j\left(\frac{t^2}{2}\right)\cot\varphi}\tilde{h}_1(\csc\varphi\cdot t)
\end{aligned}$$

$$\begin{aligned}
g_2(t) &= F_s^{-\alpha}\left[\left(F_s^{\alpha}g\right)(\omega)\right](t)\\
&= -\sqrt{\frac{2(1+j\cot\varphi)}{\pi}}e^{-j(\varphi+\frac{\pi}{2})}e^{-j\left(\frac{t^2}{2}\right)\cot\varphi}\int_0^{\infty}\left(F_s^{\alpha}g\right)(\omega)e^{-j\left(\frac{\omega^2}{2}\right)}\sin(\csc\varphi\cdot t\omega)\mathrm{d}\omega\\
&= -\sqrt{\frac{2(1+j\cot\varphi)}{\pi}}e^{-j\pi}e^{-j\left(\frac{t^2}{2}\right)\cot\varphi}\int_0^{\infty}\tilde{H}_2(\omega)\csc\omega\sin(\csc\varphi\cdot t\omega)\mathrm{d}\omega\\
&= -2\sqrt{1+j\cot\varphi}e^{-j\pi}e^{-j\left(\frac{t^2}{2}\right)\cot\varphi}\tilde{h}_2(\csc\varphi\cdot t)
\end{aligned}$$

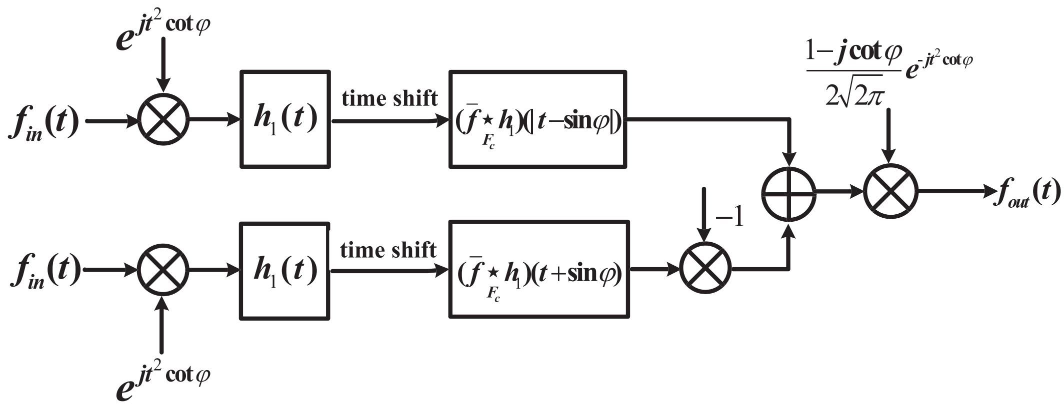

Figure 6. The method to achieve the multiplicative filters in the FRCT domain by convolution in the time domain.

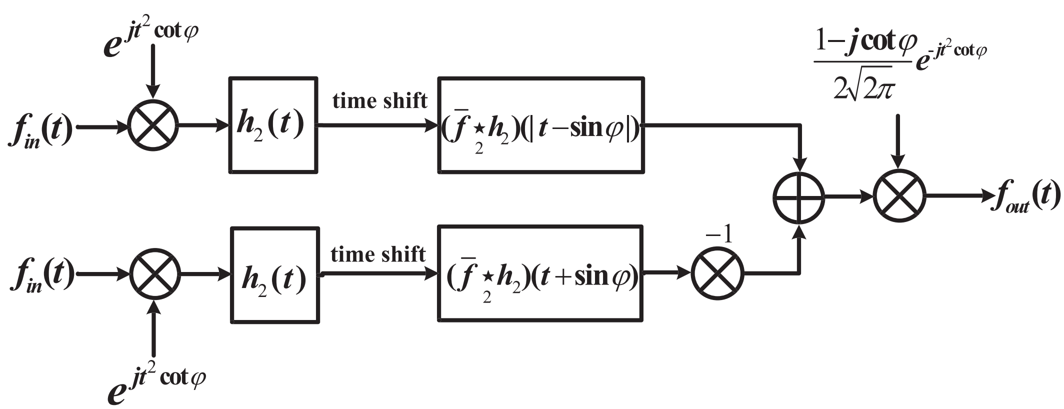

Figure 7. The method to achieve the multiplicative filters in the FRST domain by convolution in the time domain.

where $\tilde{h_1}(t)$ is the inverse FCT [27] of $\bar{H_1}(\omega)$, and $\tilde{h_2}(t)$ is the inverse FST [27] of $\bar{H_2}(\omega)$, $\bar{H_1}(\omega) = \tilde{H_1}(\omega)\sec\omega$, $\bar{H_2}(\omega) = \tilde{H_2}(\omega)\csc\omega$. Therefore we obtain

$$h_1(t) = \tilde{h_1}(\csc\varphi \cdot t), \tag{4.63}$$

$$h_2(t) = -\tilde{h_2}(\csc\varphi \cdot t) \tag{4.64}$$

Substituting Eq. (4.63), (4.64) into Eq. (4.61), (4.62), we obtain

$$\begin{aligned} f_{out}(t) = \frac{1-j\cot\varphi}{2\sqrt{2\pi}} e^{-jt^2\cot\varphi} \Big[&(f_{in}(|t-\sin\varphi|)e^{2C_{t-\sin\varphi,\varphi}}) \underset{F_c}{\star} \tilde{h_1}(|\csc\varphi\cdot t-1|) \\ &+(f_{in}(t+\sin\varphi)e^{2C_{t+\sin\varphi,\varphi}}) \underset{F_c}{\star} \tilde{h_1}(\csc\varphi\cdot t+1) \Big] \end{aligned} \tag{4.65}$$

When $f_{in}(t)$ is even signal. And

$$\begin{aligned} (f_{out}(t) = \frac{1-j\cot\varphi}{2\sqrt{2\pi}} e^{-jt^2\cot\varphi} \Big[&(f_{in}(t+\sin\varphi)e^{2C_{t+\sin\varphi,\varphi}}) \underset{2}{\star} \tilde{h_2}(\csc\varphi\cdot t+1) \\ &-(f_{in}(|t-\sin\varphi|)e^{2C_{t-\sin\varphi,\varphi}}) \underset{2}{\star} \tilde{h_2}(|\csc\varphi\cdot t-1|) \Big] \end{aligned} \tag{4.66}$$

when $f_{in}(t)$ is odd signal.

According to Theorem 3-4, it will be easy to prove that the output of filter shown in Figure 6-7 is same as that in Figure 5. This result sates that the multiplicative filter can be implemented by the FCT convolution and the FCT-FST convolution which are similar to the FT convolution, so they can be done by the FFT algorithm in the time domain.

From the above discussion, we know that the effects of the weight-function of Theorem 3-4 is only in complexity of transfer function $\widetilde{H_1}(s), \widetilde{H_2}(s)$, rather than computational complexity of convolution operation in (4.65), (4.66). Thus, the method shown in Figure 6-7 has less computational complexity than those introduced in [32, 35, 37] for processing even and odd signals. For N point of samples, the computation complexity of FFT is $O(N\log_2^N)$, meanwhile, the computation complexity of FCT or FST is one half of the FT, so the computational complexity of Eq. (4.72) and (4.73) is $O(\frac{1}{2}N\log_2^N)$, but the computation complexity of method in [32, 35, 37] is $O(N\log_2^N)$. Therefore, the implementation of the multiplicative filter based on the proposed convolution is useful for actual applications.

5. Conclusions

In this paper, based on the definition of FRCT and FRST, the convolution theorems for the FRCT and FRST, that is, Theorems 1-5, have been derived in detail, presenting the convolution theorem in the FRCT and FRST domain as well as a product between two functions in the time domain. Furthermore, Theorem 6-7 which is another version of convolution operation about Definition 1-4, is derived for analyzing multiplicative filter in the FRCT and FRST domain expediently. Finally, as an application, utilizing convolution theorems derived in this paper, the multiplicative filter in the FRCT and FRST domain has also been discussed in this paper. A practical method to achieve the multiplicative filter through convolution in the time domain is proposed. This method can be realized by classical FFT and has the same capability, but less computational complexity compared with the method achieved in the FRFT domain as well as in [32, 35, 37].

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